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DC Energy Meter

DATA SHEET

Hardware Specifications

Description	Specification
Measurement Parameters	
DC Voltage	1 Channel
DC Current	4 Channels
DC Supply	From Rectifier DC voltage (ie self-powered)
Range of Measurement Parameters	
DC Voltage	20-60 V DC
DC Current	1-100A/200A DC
Resolution of Measurement Parameters	
DC Voltage	0.1V or better
DC Current	0.2A or better
Measurement Input Method	
DC Voltage	2 pins direct input connected to rectifier output
DC Current	4 pin connectors for each channel
Measurement Accuracy	
DC Voltage	1% FSR @ 25°C or better
DC Current	1% FSR @ 25°C or better, between 5 to 95% of FSR
Temperature Coefficient	0.05%/°C
Derived Parameters	
Power	kW
Energy	kWh
Range of Derived Parameters	
Power	0-65kW
Energy	0-100000 kWh
Displayed Resolution of Derived Parameters	
Power	0.1 kW
Energy	0.1 kWh
Accuracy of Derived Parameters	
Power	1% FSR @ 25°C or better
Energy	1% FSR @ 25°C or better
Displayed Parameters	
Cumulative Energy	All 4 channels
Power	All 4 channels
Ampere	All 4 channels
Voltage	Common Rectifier Voltage

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Range of Editable Parameters	
Current Rating	1-100A
Slave ID	1 to 255
Cumulative Energy	Reset all 4 channels (facility in case of malfunction/hang up/else limit of software max value)
Channel	Enable/Disable all 4 channels
RTC	Real Time Clock Setting
Password	Change password
Front panel switches for editing input	2-4 front panel mounted membrane or tactile
Input Impedance	
Voltage	>100k ohms in Measured circuit in active state
Current	>10k ohms active state per channel
Inbuilt Protection for Rugged Telecom Applications	
Voltage	1.2 times rating and reverse polarity
Current Channel	Upto 60V
RS485 Port	Transient Above 4K V
Editable Parameters	Accessible only with password
Cumulative Energy (kWh)	Stored and incremented in memory
Isolation	Galvanic and Optical between measurement and RS485 port
Termination Connector on DCEM Back Plane	
Voltage Measurement Port and Supply Port	2 Pin Male/Female Pluggable for overall voltage
Current measurement port	4 Pin Male/Female Pluggable for voltage per channel
Serial port RS485	3 pin Male/Female Pluggable
Contact Material/Solder Pin	Brass with Tin Plating
Marking	Non Erasable ink with channel input markings, voltage markings (+/-) & serial port RS485 markings
Internal Details of DCEM-Critical Components /Software Inputs	
Front screen display	16 x 4 alphanumeric LCD with backlight
Memory	Epsom 64k
Memory Life	5 years or more without corruption
RTC-Real Time Clock	High Accuracy with Button Battery Backup
Update of Display Readings	Approx 1 sec on all 4 channels simultaneously
Update of DCEM	Every 0.1 kWh increment
Power Consumption	Approx 4 watts
Environmental and General Specifications	
Operating Temp	0-55°C
Storage Temp	0-80°C
Operating Humidity	0-80% RH non condensing
Casing Material	Virgin Polycarbonate on all sides, heat resistant grade
Size of DCEM	Din rail mount 100 x 100 x 60 mm
Colour	White/Grey
Communication Serial Port Specifications	
Type	RS 485, 3 wire communication port
Protocol	Modbus RTU
Settings-1	Default Slave ID 50, Function Code 3,4,5,6, Baud Rate 9600
Settings-2	1 Stop Bit, No Parity
Interface	DX-, DX+, GND
Isolation	4KV

Ordering Information

Item Code	INV-METER-0065

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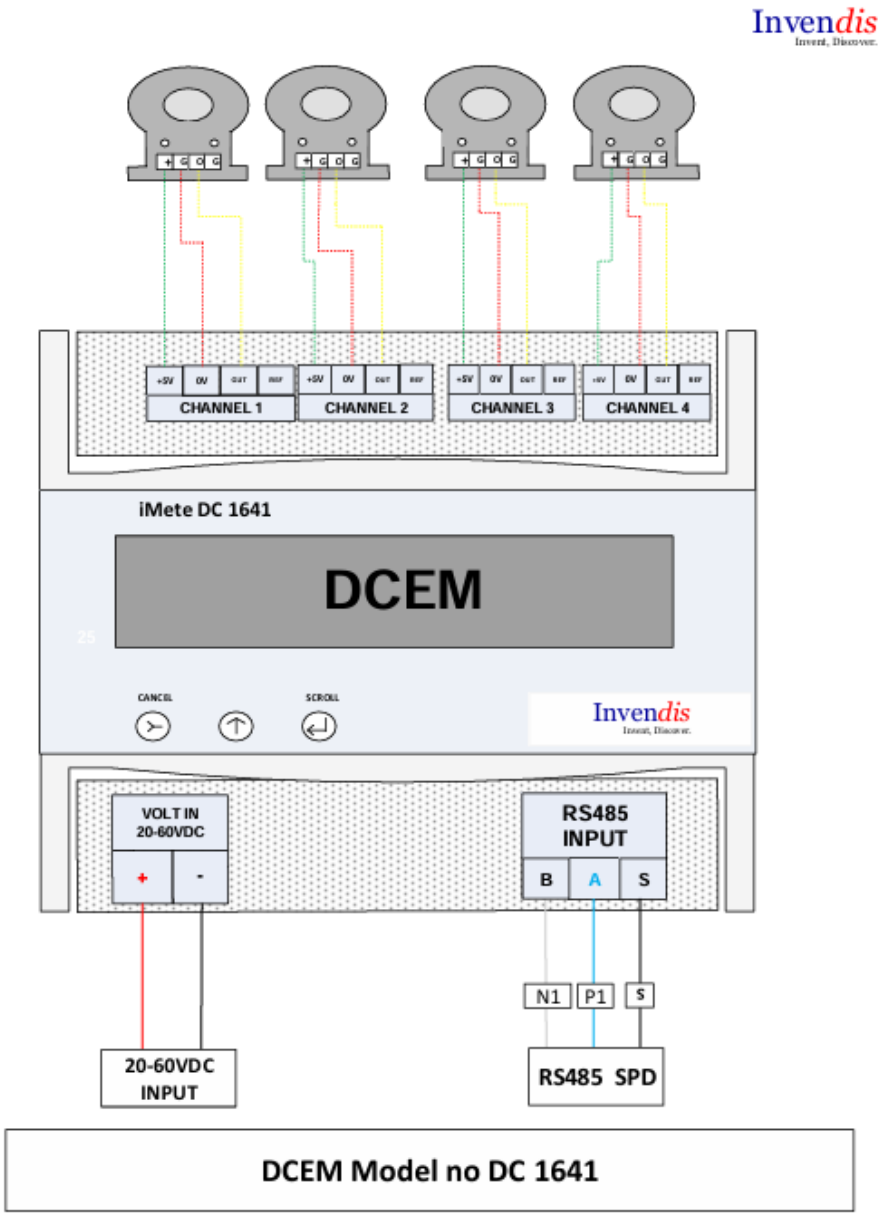
CONNECTION DIAGRAM



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